LAB 05 - PROMPTS

Indicator Stage: Pine Script v6 Indicators and their Python Twins

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab05-pine-indicators/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Add the trend and bands indicator (A3, K3)

Prompt - copy into your AI coding assistant
Explain labs_fullcode/Lab05-pine-indicators/lcc_trend_bands.pine line by line for a finance professional: the inputs, ta.ema, ta.bb, ta.crossover / ta.crossunder, plotshape, barcolor and alertcondition. Which values will the Data Window show for each bar?

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Check: EMA 20 (blue), EMA 50 (orange), shaded Bollinger Bands and green/red crossover triangles on the price pane.

Step 2: Add the RSI pane (A3, K3)

Do this (no prompt needed)
Create another indicator from Lab05-pine-indicators/lcc_rsi.pine, save it as 'LCC RSI' and add it.

Check: RSI(14) in its own pane with 70/30 lines and shaded overbought/oversold zones.

Step 3: Record the TradingView values (A3)

Do this (no prompt needed)
Move the cursor to 31 December 2025, open the Data Window and copy close, EMA fast, EMA slow, BB upper, middle, lower and RSI into Lab05-pine-indicators/tradingview_indicator_values.csv (overwrite the example row).

Check: Default (unadjusted) chart: close 271.86, EMA 20 about 273.98, EMA 50 about 269.60, BB 282.35 / 275.47 / 268.60, RSI about 45.64.

Step 4: Build the Pine twins (K3)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/indicators.py with Python twins of the Pine Script v6 built-ins, implemented by calling labs_fullcode/common/indicators.py: ta_sma(src, length), ta_ema(src, length) (alpha 2/(n+1), adjust=False), ta_rsi(src, length=14) (Wilder smoothing), ta_bb(src, length=20, mult=2.0) returning (middle, upper, lower) with the population standard deviation, ta_atr(bars, length=14), ta_macd(src, 12, 26, 9) returning (macd, signal, hist), ta_crossover(a, b) = a > b and a[1] <= b[1], ta_crossunder(a, b). Then lcc_trend_bands(bars, fast=20, slow=50, bb_len=20, bb_mult=2.0) returning close, ema_fast, ema_slow, bb_middle, bb_upper, bb_lower, bull_cross, bear_cross exactly like lcc_trend_bands.pine, lcc_rsi(bars, 14, 70, 30) returning rsi, alert_oversold and alert_overbought like lcc_rsi.pine, and indicator_table(bars) joining both. Every value at bar t may use data up to bar t only. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: indicator_table(load_bars('AAPL','2018-01-01','2025-12-31')).iloc[-1] shows ema_fast 273.23 and rsi 45.69.

Step 5: Build the cross-check run (A3, K3)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/indicators.py add run(symbol='AAPL', date='2025-12-31', tv_file=None) for `python -m pipeline indicators --symbol --date --tv`: compute indicator_table on the ADJUSTED research bars (from config data.start) and on the RAW reference bars from pipeline.validate.reference_bars (TradingView's default view); print a snapshot table (close, ema_fast, ema_slow, bb_upper, bb_middle, bb_lower, rsi) with columns Python ADJUSTED, Python RAW (TV default), TradingView (you) from Lab05-pine-indicators/tradingview_indicator_values.csv and 'diff vs RAW' in %; the trend, band and RSI reading; the number of bullish/bearish crosses and RSI alert events since 2024-06-01; PASS if the largest gap is <= 0.5%. Save _outputs/Lab05_<SYMBOL>_indicators.csv and chart Lab05_pine_preview.png (price, EMAs, bands, cross markers, RSI pane). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline indicators --help shows --symbol, --date and --tv.

Step 6: Run the cross-check (A3, K3)

Do this (no prompt needed)
Run: python -m pipeline indicators --symbol AAPL --date 2025-12-31 (same as: python3 Lab05-pine-indicators/indicators_stage.py --symbol AAPL).

Check: ADJUSTED close 271.12, EMA 20/50 273.23 / 268.80, BB 281.58 / 274.72 / 267.87, RSI 45.69; RAW 271.86, 273.98 / 269.60, 282.35 / 275.47 / 268.60, RSI 45.64; largest gap to TradingView 0.006% -> PASS; trend UP; since 2024-06-01: 2 bullish and 2 bearish EMA crosses, 3 RSI-oversold and 15 RSI-overbought alert events.

Step 7: Change the inputs and count the extra trading (K3, K7)

Prompt - copy into your AI coding assistant
In a Python shell started in labs, use pipeline.indicators.lcc_trend_bands on pipeline.ingest.load_bars('AAPL', '2018-01-01', '2025-12-31') with fast/slow 20/50 and then 10/30, count bull_cross and bear_cross since 2024-06-01 for each, and estimate the extra trading cost of the faster setting at 3 bps per round trip on USD 100,000. Print a small table.

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Check: Since 2024-06-01: EMA 20/50 = 2 bullish / 2 bearish crosses; EMA 10/30 = 7 / 7 - about 3.5 times as many trades, so cost rises with signal frequency.

Step 8: Create the alert the automation stage will use (A3, K7)

Do this (no prompt needed)
Restore EMA 20/50. Click Alert, choose Condition: LCC Trend > 'EMA bullish cross', trigger 'Once per bar close', and save. This alert type is what Lab 17 connects to the pipeline webhook.

Check: The alert appears in the Alerts panel, active, on NASDAQ:AAPL.

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Run it

Step	Command / file
Run Lab 05	python3 Lab05-pine-indicators/indicators_stage.py (same as: python -m pipeline indicators)
The complete code	pipeline/indicators.py (reference copy: Lab05-pine-indicators/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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